• **DRIVEABILITY—ENGINE LACKS POWER WHEN OPERATING IN HIGH AMBIENT TEMPERATURES ABOVE 38C (100F)—VEHICLES EQUIPPED WITH 5.4L ENGINE ONLY** • **ENGINE—5.4L—ENGINE LACKS POWER WHEN OPERATING IN HIGH AMBIENT TEMPERATURES ABOVE 38C (100F)**	**Article No. 02-12-3**

FORD: 2000-2002 EXPEDITION, F-150

ISSUE

Some vehicles equipped with the 5.4L-2V engine may exhibit a lack of power condition when operating in ambient temperatures above 38°C (100 °F). This may be caused by the Powertrain Control Module (PCM) calibration which retards the ignition timing to prevent detonation when the Inlet Air Temperature (IAT) reaches 60°C (140 °F) or above. This may be more pronounced during stop-and-go driving.

ACTION

Verify the condition. Reprogram the PCM with the latest calibration and on 2000 and 2001 F-150's, install body plugs behind Coolant Overflow Bottle if missing. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

For 2000-2002 F-150's and Expeditions equipped with a 5.4L 2V engine, if a lack of power occurs only when the IAT is above 60°C (140°F), reprogram the PCM to the latest level calibration listed in this TSB.

For 2000 and 2001 F150's, install Body Plugs (383181-S and N807093-S) behind the Coolant Overflow Bottle. Refer to Figure 1 for location and plug button installation. These plugs will reduce the engine compartment heat moving into the air cleaner inlet area.

Body Plug Installation

1. Remove air cleaner inlet. Refer to Section 303-12.
2. Remove jack handle.
3. Remove upper radiator sight shield.
4. Disconnect radiator overflow hose from radiator to degas bottle.
5. Position degas bottle away from body plug hole area.
6. Install body plugs in location shown in illustration.
7. Position back degas bottle Torque to 9 N•m (80 Lb-in).
8. Connect radiator overflow hose.
9. Install upper radiator sight shield.
10. Install jack handle.
11. Install air cleaner inlet. Refer to Section 303-12.

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PCM CALIBRATION INFORMATION						
Application	Old Part Number (-12A650-)	Tear Tag	New Part Number (-12A650-)	Old Calibration	New Calibration	NGS/WDS Qualifier
2000 F-150, 5.4L, 4R70W, Fed/Alt/Can	YU2F-AGA	JSQ0	YU2Z-AGB	0F514A0A06	0F514A0A06	WDS B19A.5 Release or Later
2000 F-150 5.4L 4R70W Fed/Alt/Can (Non LEV)	YU2F-AHA	RZM0	YU2Z-AHB	0F514N0A06	0F514N0A06	WDS B19A.5 Release or Later
2000 F-150, 5.4L Calif, 4R100	YL3F-LE	CPE4	YU7Z-BHA	0F514D0B11	0F514D0B11	WDS B19A.5 Release or Later
2000 F-150, 5.4L Calif, 4R100	YL3F-ATE	SNL4	YU7Z-BZA	0F514E0B13	0F514E0B13	WDS B19A.5 Release or Later
2000/2001 F-150, 5.4 49S/Alt/Can/Mex, 4R70W	YL3F-AUD	UAT3	YU7Z-CAA	0F514F0A12	0F514F0A12	WDS B19A.5 Release or Later
2000 F-150, 5.4, 4R100 Fed/Alt/Can, Heavy Duty	YL3F-BCB	GMH1	YU7Z-DNA	0F514G0A11	0F514G0A11	WDS B19A.5 Release or Later
2000 F-150, 5.4 Fed/Alt/Can, 4R100	YL3F-BDB	JBR1	YU7Z-DPA	0F514H0A11	0F514H0A11	WDS B19A.5 Release or Later
2000 F-150, 5.4L, 4R100, Calif.	YC3F-PE	NCG4	YU7Z-DRA	0F514M0B11	0F514M0B11	WDS B19A.5 Release or Later
2001 F-150 5.4L, Calif, Base/Harley, 4R70W	1U7A-BKA	KKM0	1U7Z-BKB	1F514E0B11	1F514E0B11	WDS B19A.5 Release or Later
2001 F-150, 5.4L 4R70W Fed/Base/Harley	1U7A-BLA	LDN0	1U7Z-BLB	1F514F0A11	1F514F0A11	WDS B19A.5 Release or Later
2001 F-150, 5.4L, Base, Calif, 4R100	1L3F-TD	PTJ3	1U7Z-BYA	1F514D0B10	1F514D0B10	WDS B19A.5 Release or Later
2001 F-150, 5.4L, Fed, 4R100	1L3F-ATD	ILM3	1U7Z-CDA	1F514G0A10	1F514G0A10	WDS B19A.5 Release or Later
2001 F-150, 5.4L, Fed, 4R100	1L3F-AUD	CRG3	1U7Z-CEA	1F514H0A10	1F514H0A10	WDS B19A.5 Release or Later
2001 F-150, 5.4L, Calif, 4R100	1L3F-UD	LBJ3	1U7Z-CFA	1F514M0B10	1F514M0B10	WDS B19A.5 Release or Later
2002 F-150, 5.4 Base, Calif, 4R100 Bi-Fuel	2L3A-TC	TGR2	2U7Z-ANA	2F514D0B06	2F514D0B06	WDS B19A.5 Release or Later
2002 F-150, 5.4 Base, Calif, 4R70W	2L3A-VD	VMF3	2U7Z-APA	2F514E0B07	2F514E0B07	WDS B19A.5 Release or Later
2002 F-150, 5.4 Base, Fed, 4R70W	2L3A-WD	WUQ3	2U7Z-ARA	2F514F0A07	2F514F0A07	WDS B19A.5 Release or Later
2002 F-150, 5.4 Fed, 4R100	2L3A-YC	RPW2	2U7Z-ASA	2F514H0A06	2F514H0A06	WDS B19A.5 Release or Later
2002 F-150, 5.4, Calif, 4R70W	2C3A-CB	BGR1	2C3Z-CCC	2F514A0B05	2F514A0B06	WDS B19A.5 Release or Later
2002 F-150, 5.4, Calif.	2L3A-ATB	BWJ1	2L3Z-ATC	2F514E0B16	2F514E0B17	WDS B19A.5 Release or Later
2002 F-150, 5.4, 49S	2L3A-AUB	UEV1	2L3Z-AUC	2F514F0A16	2F514F0A17	WDS B19A.5 Release or Later
2000 Expedition, 5.4L 2V 50S/Alt/Can 4R100	YL1F-BE	ZUT4	YU7Z-AYA	0B314B0511	0B314B0511	WDS B19A.5 Release or Later
2001 Expedition, 5.4L, Fed/Calif/LEV, 4R100	1L1F-CC	PWS2	1U7Z-BXA	1B314B0510	1B314B0510	WDS B19A.5 Release or Later
2002 Expedition, 5.4L 4R100, Less Mexico	2L1A-FB	AKD1	2U7Z-ALA	2B314B0505	2B314B0505	WDS B19A.5 Release or Later
2002 Expedition, 5.4L 2V A/T Mexico	2L1A-JA	JJP0	2U7Z-AMA	2B314B0M00	2B314B0M00	WDS B19A.5 Release or Later

Application	Old Part Number (-12A650-)	Tear Tag	New Part Number (-12A650-)	Old Calibration	New Calibration	NGS/WDS Qualifier

PART NUMBER	PART NAME
383181-S	Body Plug
N807093-S	Body Plug

Obtain an Authorized Modifications Decal (STAT 5941-2 - obtainable through the PDC) and list the date, dealer number, and summary of modifications performed. Select a prominent place adjacent to the Vehicle Emission Control Information Decal suitable for installing the Authorized Modifications Decal. Clean the area, install the decal, and cover it with a clear plastic decal shield.

AUTHORIZED MODIFICATIONS

THE FOLLOWING MODIFICATIONS HAVE BEEN MADE:

Reprogrammed Powertrain Control Module (PCM) Per TSB 02-12-3

THESE MODIFICATIONS HAVE BEEN APPROVED, AS APPROPRIATE, BY EPA AND CARB.

DEALER NUMBER: DATE:

CHANGE AUTHORITY:

FPS 8262 9/78 FORD MOTOR COMPANY PRINTED IN U.S.A.

OTHER APPLICABLE ARTICLES: NONE

WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage And Emissions Warranty Coverage

OPERATION	DESCRIPTION	TIME
021203A	Reprogram Powertrain Control Module: 2000, 2001, 2002 F-150, F-150 Super Crew, Expedition	0.6 Hr.
021203B	Reprogram Powertrain Control Module And Install Body Plugs If Missing: 2000, 2001 F-150, F-150 Super Crew	0.9 Hr.

DEALER CODING

	CONDITION CODE
BASIC PART NO.	
RECALEM	42

OASIS CODES: 614000, 614500, 614600, 622000

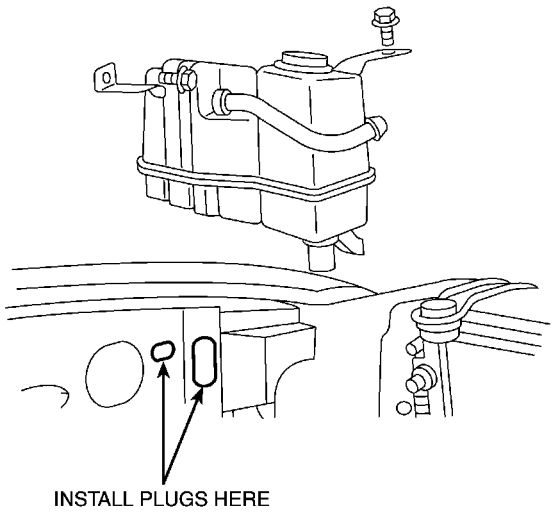


Figure 1 - Article 02-12-3